

Cohen-Macaulay
complexes

Face ring of a simplicial complex with vertex set $[n]$.

$$k[\Delta] = k[x_1, \dots, x_n] / I_\Delta$$

$$I_\Delta = \left\langle x^s : s \notin \Delta \right\rangle$$

" $\prod_{i \in s} x_i$

Hilbert series $\Rightarrow$

$$H(k[\Delta], t) = \sum_{i=0}^{\infty} \dim k[\Delta]_i t^i$$

$$= \frac{h_0 + h_1 t + \dots + h_{d+1} t^{d+1}}{(1-t)^{d+1}}$$

Throughout, let A be a finitely generated graded K -algebra.

A **regular sequence** on A is a sequence

$\theta_1, \dots, \theta_n \in A$ such that for all $1 \leq i \leq n$,

θ_i is a nonzerodivisor in $A / \langle \theta_1, \dots, \theta_{i-1} \rangle$

If $\theta_i x = 0$, then

$$x = 0$$

Suppose $\theta_1, \dots, \theta_n$ is a regular sequence of **degree 1** elements.

Multiplication by θ_1 is an injective map $A \rightarrow A$

$$H(\theta_1 A, t) = t H(A, t)$$

$$\begin{aligned} H(A/\theta_1 A, t) &= H(A, t) - H(\theta_1 A, t) \\ &= (1-t) H(A, t) \end{aligned}$$

By induction

$$H(A / \langle \theta_1, \dots, \theta_n \rangle, t) = (1-t)^n H(A, t)$$

$$\text{If } A = k[\Delta], \quad n = d+1$$

$$H(A / \langle \theta_1, \dots, \theta_n \rangle, t) = h_0 + h_1 t + h_2 t^2 + \dots + h_{d+1} t^{d+1}$$

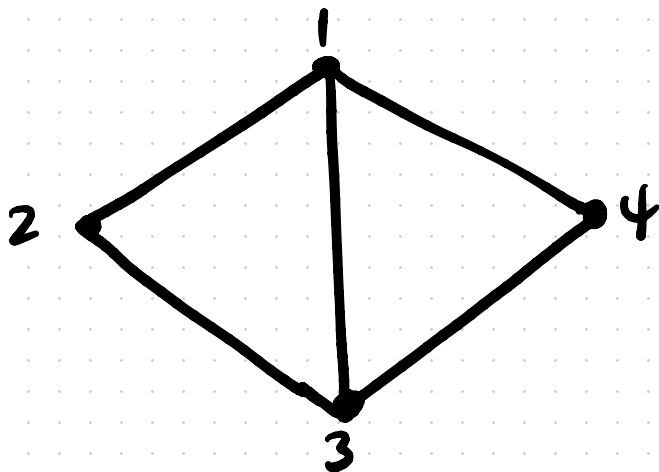
$$\Rightarrow h_i \geq 0$$

A homogeneous system of parameters (hsop) for A is a sequence $\theta_1, \dots, \theta_n \in A$ of homogeneous elements such that $A / \langle \theta_1, \dots, \theta_n \rangle$ is a finite-dimensional vector space, and n is minimal among all such sequences.

A linear system of parameters (lsop) is a hsop of degree 1 elements.

Thm (Noether Normalization Lemma): Let A be a graded k -algebra which is finitely generated by degree 1 elements. Assume k is infinite. Then A has a lso.

Prop (Stanley): Let Δ be a d -dim simplicial complex. Then $\theta_1, \dots, \theta_n \in k[\Delta]$, is a lso iff $n = d+1$ and for all $\sigma \in \Delta$, the restriction of $\theta_1, \dots, \theta_n$ to σ linearly spans $\text{span}_k(x_i : i \in \sigma)$.



$$\theta_1 = x_1$$

$$\theta_2 = x_3$$

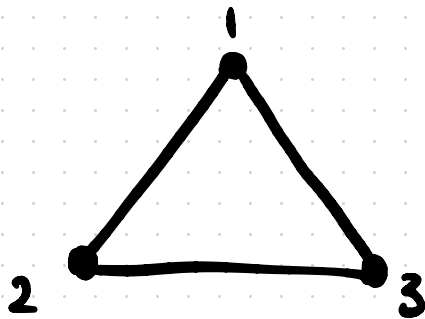
$$\theta_3 = x_2 + x_4$$

$$k[\Delta] / \langle x_1, x_2 + x_4, x_3 \rangle$$

$$\cong k[x] / \langle x^2 \rangle$$

$$H(\dots, t) = 1 + t$$

$$\Delta = 2^{[3]} / [3]$$



$$\theta_1 = a_{11}x_1 + a_{12}x_2 + a_{13}x_3$$

$$\theta_2 = a_{21}x_1 + a_{22}x_2 + a_{23}x_3$$

will be a bop for any
generic a_{ij} .

$$A / \langle \theta_1, \theta_2 \rangle$$

$$\cong k[x] / \langle x^3 \rangle$$

$$H(\dots, t) = 1 + t + t^2$$

We say A is **Cohen-Macaulay** if it has a hsop which is a regular sequence.

If so, then **every** hsop is a regular sequence.

We call a simplicial complex Δ **Cohen-Macaulay** if $k[\Delta]$ is Cohen-Macaulay.

Thm (Reisner's criterion): Δ is Cohen-Macaulay iff for all $\sigma \in \Delta$ (including $\sigma = \emptyset$),

$$\tilde{H}_i(\text{link}_{\Delta}\sigma) = 0 \quad \forall i < \dim(\text{link}_{\Delta}\sigma)$$

In particular, if Δ is a simplicial ball or sphere, then it is Cohen-Macaulay.

Let Δ be Cohen-Macaulay of dim d .

Assuming k is infinite, there is a isop $\theta_1, \dots, \theta_{d+1}$ for $k[\Delta]$.

Since $k[\Delta]$ is C-M, $\theta_1, \dots, \theta_{d+1}$ is a regular sequence.

Therefore,

$$H(k[\Delta] / \langle \theta_1, \dots, \theta_{d+1} \rangle, t) = h_0 + h_1 t + \dots + h_{d+1} t^{d+1}.$$

Thm (Stanley's upper bound theorem): For any simplicial $(d-1)$ -sphere Δ with n vertices

$$h_k(\Delta) \leq \binom{n-d+k-1}{k}$$

Proof:

h_k is the dimension of the k^{th} -graded component of

$B = \mathbb{K}[\Delta] / \langle \theta_1, \dots, \theta_d \rangle$, where $\theta_1, \dots, \theta_d$ is an isop

B is generated in degree 1, and $\dim B_1 = n-d$.

Suppose $y_1, \dots, y_{n-d}$ is a linear basis for B_1 .

Every element of B can be written as a polynomial in $y_1, \dots, y_{n-d}$.

Therefore the monomials in $y_1, \dots, y_{n-d}$ of degree k span B_k . The number of such monomials is $\binom{n-d+k-1}{k}$.

Therefore

$$h_k = \dim B_k \leq \binom{n-d+k-1}{k}$$